Defect-Aware Multi-Die Image Fusion for Semiconductor Reference Reconstruction

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Proposed direction · not yet built or validated

RESEARCH QUESTION (HYPOTHESIS) *Can defect-aware fusion of aligned, same-design die images produce a more faithful candidate reference than single-image and conventional fusion baselines, while reliably identifying regions that lack sufficient evidence?*

1 Problem and idea

Inspection often compares a die image with a reference; if that reference is itself a real image, its own defects and noise enter the comparison. Same-design dies should show the same pattern while random defects fall in different places, so 4–5 images hold **complementary evidence** (Fig. 1A). Because they are captured at slightly different positions and angles, they must first be **registered** into one frame (B). The project estimates which regions are reliable (C), rebuilds the reference from reliable evidence only (D) and flags regions without such evidence (E). The output is a **digital reference image**, not a fabricated or repaired die; looking clean does not imply electrical function.

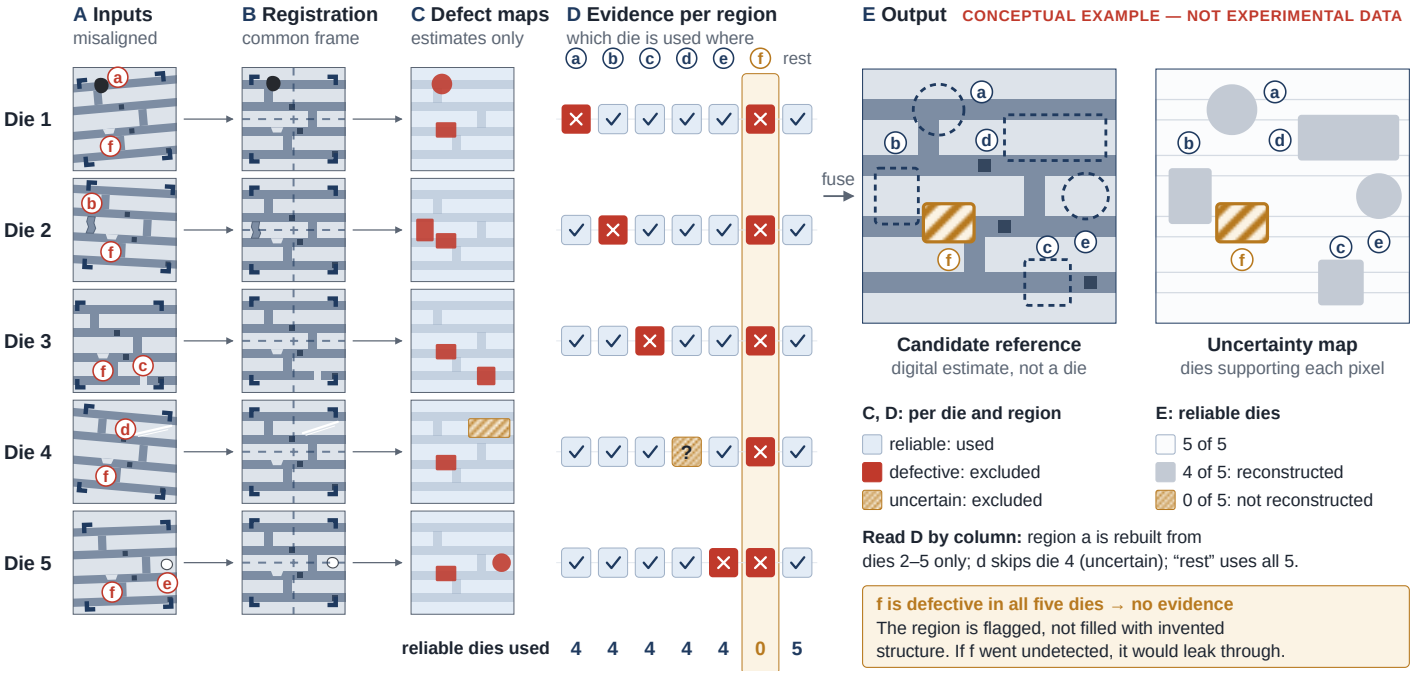


Fig. 1 · Conceptual example, not experimental data. One synthetic layout in every panel; the maps in C are estimates, not ground-truth labels.

2 Objectives

O1 Register 4–5 same-design images; measure alignment error.

O2 Compare with single, mean, median, anomaly-masked fusion.

O3 Estimate defective, reliable and uncertain regions per image.

O4 Evaluate on synthetic groups and, if obtainable, real pairs.

3 Research gap

Defect classification [4], anomaly segmentation [1, 2], registration and image fusion are established, not new. Segmentation labels *what is wrong* in one image; this project estimates *what the pattern should be*, and where that estimate is unsupported. What is tested is the combination of multi-die registration, region-level reliability estimation, evidence-weighted reconstruction and explicit uncertainty for under-observed regions; its novelty must be confirmed by a systematic literature and patent review.

4 Evaluation

Metrics: **fidelity** (PSNR, SSIM against the true clean reference), **structure** (edge, line-width error), **defect leakage**, **registration** error and **uncertainty** quality. The approach is useful only if, on held-out groups, it leaks fewer defects than mean or median fusion at similar fidelity, beats a single-image reference, and concentrates its uncertainty flags on truly unrecoverable regions; negative results will be reported.

5 Data feasibility

Dataset	Content (provider description)	Annotation	Groups / clean ref.?	Licence
Carinthia [3]	4,591 SEM images; one layer, unstructured wafers	Image-level class (6 classes)	No / No	not verified
Carinthia-S [3]	Same images, enhanced release	Binary defect masks + class	No / No	CC BY 4.0 (unverified)
MVTec AD [1]	5,354 images, 15 object/texture categories	Pixel masks; defect-free train set	No / No (unpaired)	CC BY-NC-SA 4.0
MVTec AD 2 [2]	>8,000 images, 8 scenarios, varied lighting	Pixel masks (private test withheld)	No / No (unpaired)	CC BY-NC-SA 4.0

"Groups" means same-design image groups with complementary defects; "clean ref." means a defect-free reference of the same die. None provides multi-die reconstruction ground truth, so the project needs a **controlled synthetic benchmark** (known shifts, noise and inserted real defects give an exact reference) and **real same-design groups** via industry collaboration to test transfer.

[1] P. Bergmann, M. Fauser, D. Sattlegger, C. Steger, "MVTec AD — A comprehensive real-world dataset for unsupervised anomaly detection," *CVPR*, 2019, pp. 9584–9592. doi:10.1109/CVPR.2019.00982 · dataset

[2] L. Heckler-Kram, J.-H. Neudeck, U. Scheler, R. König, C. Steger, "The MVTec AD 2 dataset: Advanced scenarios for unsupervised anomaly detection," *IJCV*, 2026. doi:10.1007/s11263-026-02743-0 · arXiv · dataset

[3] C. Kofler, S. Strauß, A. Zernig *et al.*, "Carinthia dataset," v2, Zenodo, 2024, doi:10.5281/zenodo.10715190; C. Kofler, V. Hasić, "Carinthia-S," Zenodo, 2025, doi:10.5281/zenodo.16895427.

[4] K. Taha, "Observational and experimental insights into machine learning-based defect classification in wafers," *J. Intell. Manuf.*, Springer. doi:10.1007/s10845-024-02521-0